Adarsh Dwivedi

Machine Learning Engineer

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Final-year CS engineer who ships models with their error cost printed next to their accuracy. Built a fraud-ring detector at 0.7292 precision against a 0.2242 base rate and reported the customers swept in per fraudster caught, benchmarked 8 model families against a 0.291% label rate, and validated a credit model against 117,454 real borrowers to close a circularity gap. Placed 11th of 93 on a national leaderboard.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**
AI Engineer (Agentic AI) Intern Gurugram, India

- Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.
- Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.
- Built capability in five layers on a multi-agent travel planner, from a single agent to a supervisor with three specialists, a human approval gate, real browser tools over MCP and serving through Aegra, proving each layer before starting the next.

PROJECTS

Orbweaver, Fraud Ring Detection in Transaction Graphs | Live | GitHub Python, igraph, Densest Subgraph, Knapsack

- Promotion abuse is a group behaviour, so scoring accounts one at a time misses it. Built a multi-relation account graph with edges weighted by entity rarity times measured fraud lift, fitted on training accounts only.
- Scored and pruned first, then extracted rings by greedy densest-subgraph peeling with a proved one-half approximation bound, reaching 0.7292 ring precision against a 0.2242 base rate on a 35.7M-edge graph.
- Attached the cost to every number: 0.371 genuine customers swept in per fraudster caught, an exact knapsack over reviewer minutes sizing the queue to analyst capacity, and 36 dated failures published.

MargaDrishti, Spatio-Temporal ML for Urban Road-Capacity Loss | Live | GitHub Python, H3, LightGBM, 8 Model Families

- Modelled Bengaluru road-capacity loss over 298k violations on one H3-by-hourly substrate, with strictly causal lag features at t-1h, t-24h and t-168h so nothing leaks from the future.
- Engineered spatial features as H3 at three resolutions with k-ring neighbour aggregates and target-encoded stations, plus OSM built-environment signals including junction degree and edge betweenness.

KrishiMitra, Crop Intelligence Platform for Indian Smallholders *(Deployed, 12 Indian languages)* **| Live | GitHub** CatBoost, FastAPI, Next.js, FAO-56

- A model trained on soil chemistry alone will confidently recommend a crop a district has never grown. Cross-checked every CatBoost recommendation against five years of government district returns.
- Detected leaf disease at 93.75% across 10,162 labelled images over 13 classes, validated on field-disjoint splits so no field appears in both train and test.

CreditSetu, Explainable Credit Scoring for Thin-File Borrowers *(Published on PyPI)* **| Live | GitHub** LightGBM, SHAP, ruptures, Python

- Thin-file borrowers have no bureau history, so scored them from behavioural signals instead: income entropy, rent consistency, bounce history and EMI burden, with LightGBM handling the sparsity natively.
- Detected life events straight from cash flow with ruptures PELT change-point detection, surfacing EMI closures and income step-ups a static snapshot would miss.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Pandas, NumPy, Feature Engineering, Cross-Validation, Calibration, SHAP, A/B Testing, Statistics

Deep Learning: PyTorch, TensorFlow/Keras, CNNs, ResNet50, VGG16, EfficientNetV2, CNN-LSTM, Transformers, Transfer Learning, Ordinal Regression, Grad-CAM, OpenCV

Generative AI: LangGraph, LangChain, RAG, Prompt Engineering, Vector Search, Embeddings, Evaluation and Calibration, Amazon Bedrock, Gemini, OpenAI, Groq, Ollama

Backend and Deployment: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

ACHIEVEMENTS

- SWITCH Energy-X (India): 11th of 93 on the private leaderboard, recovering a hidden physical law from 500,000 unlabeled, 26%-missing sensor rows.
- KSP Datathon 2026 finalist: reached Level 2 with Kadi, a crime analytics platform built end to end over 50 days and deployed on Zoho Catalyst for the Karnataka State Police.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**
B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.